

Multimodal Model Fusion with Ultrasound Images and Text for Breast Lesion Analysis

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Abstract—Breast cancer is one of the leading causes of death among women worldwide, underscoring the critical importance of early detection for improving survival rates. This study explores the application of machine learning algorithms to detect breast cancer using the Breast Cancer Wisconsin (Diagnostic) dataset, which contains features derived from fine needle aspiration (FNA) tumor images. We evaluate and compare the performance of three widely-used classification models: Support Vector Machine (SVM), Random Forest, and Logistic Regression. The models were assessed using standard metrics, including accuracy, precision, recall, and F1-score. Our findings indicate that the SVM model outperforms the other models, achieving the highest accuracy and demonstrating strong reliability for distinguishing between malignant and benign tumors. Additionally, the Random Forest model provides valuable insights into feature importance, which could aid clinicians in identifying key tumor characteristics. This research underscores the potential of machine learning in breast cancer diagnosis, with implications for both clinical practice and future research into more advanced models and datasets.

Index Terms—breast cancer, machine learning, classification, early diagnosis, SVM, Random Forest, Logistic Regression.

I. INTRODUCTION

BREAST cancer is a major global health concern, being one of the most commonly diagnosed cancers among women. According to the World Health Organization (WHO), breast cancer accounts for nearly 25% of all cancer cases worldwide, with millions of new cases reported annually. Early detection is crucial, as it significantly increases the chances of successful treatment and improves patient survival rates. Traditionally, methods such as clinical examinations, mammography, and biopsy have been used to detect and diagnose breast cancer. While these techniques are effective, they often face limitations such as high costs, variability in results, and dependency on clinician experience.

In recent years, machine learning (ML) has emerged as a powerful tool in medical diagnostics, offering the potential for more accurate, efficient, and automated decision-making. ML models can analyze large volumes of complex data, such as medical images and clinical records, to identify patterns that may be difficult for human experts to detect. In particular, classification models can assist in categorizing tumors as malignant or benign, helping clinicians make informed decisions about patient care.

This study investigates the use of machine learning algorithms for breast cancer detection using the Breast Cancer

Wisconsin (Diagnostic) dataset. This dataset consists of tumor features extracted from fine needle aspiration (FNA) images of breast tumors, along with labeled data indicating whether the tumors are malignant or benign. The focus of this paper is to evaluate and compare the performance of three machine learning models: Support Vector Machine (SVM), Random Forest, and Logistic Regression. By assessing the accuracy, precision, recall, and F1-score of each model, we aim to determine which algorithm provides the most reliable and efficient solution for early breast cancer detection.

II. LITERATURE REVIEW

Machine learning (ML) has become an essential tool in medical diagnostics, particularly in breast cancer detection. Leveraging multimodal data—such as combining imaging data with clinical records—has demonstrated significant potential in improving diagnostic accuracy. This section explores prior research on the datasets, methodologies, and fusion strategies relevant to this study.

The Breast-Lesions-USG dataset, as curated by Pawłowska et al. (2024), provides a benchmark for ultrasound-based breast lesion analysis. This dataset, consisting of annotated ultrasound scans and corresponding clinical data, has been widely used in studies that combine image and text data for diagnostic purposes. Its design facilitates the development and testing of machine learning models that aim to integrate multimodal inputs for improved classification performance [1].

Pereira et al. (2023) conducted a comparative study of early and late fusion strategies for binary classification problems involving multimodal data. Their findings suggest that early fusion—where features from different modalities are combined at the feature extraction stage—can effectively capture interactions between modalities, leading to improved prediction accuracy. Late fusion, by contrast, involves combining the outputs of independent models trained on separate modalities, offering simplicity and flexibility but potentially missing intricate relationships between features [2].

The CLIP (Contrastive Language-Image Pretraining) model, introduced by OpenAI (2024), has further expanded the capabilities of multimodal learning by connecting text and images through a unified representation space. While CLIP has primarily been applied in natural language processing and computer vision tasks, its principles have influenced the design of early and late fusion strategies for medical applications.

These methods aim to balance the strengths of each modality while addressing limitations such as overfitting or the lack of interpretability in combined models [3].

In their analysis of fusion methods, Boulahia et al. (2021) highlighted the effectiveness of early, intermediate, and late fusion strategies in robust multimodal classification tasks. Their research emphasized the importance of dataset quality and the careful integration of features from diverse modalities. The study noted that while early fusion may capture more nuanced relationships, late fusion allows for greater flexibility in handling varying data types or temporal sequences, making it a viable choice in complex medical scenarios [4].

Building on these prior works, this study seeks to evaluate the performance of early and late fusion strategies for breast cancer detection using the Breast-Lesions-USG dataset. By comparing the effectiveness of image-only, text-only, early fusion, and late fusion models, we aim to identify optimal approaches for integrating multimodal data in diagnostic workflows.

III. METHODOLOGY

A. Dataset Overview

This study utilizes the Breast-Lesions-USG dataset, a curated resource developed by Pawłowska et al. (2024). The dataset comprises 256 annotated ultrasound images of breast lesions, paired with corresponding clinical records. Each case includes tumor-level labels (e.g., benign or malignant) and patient-level details such as age and symptoms. Designed for research in breast cancer detection, this dataset facilitates the exploration of multimodal machine learning approaches by combining imaging and text-based clinical data [1].

B. Data Preprocessing

To prepare the dataset for multimodal machine learning models, the following preprocessing steps were conducted:

- **Handling Missing Clinical Data:** Missing values, particularly in patient age, were addressed using two strategies inspired by Pereira et al. (2023):
 - Replacing missing age values with the mean age, which provided completeness but introduced potential noise.
 - Removing records with missing data entirely to enhance model generalization and avoid potential overfitting [2].
- **Image Data Processing:**
 - Ultrasound images were resized to a consistent input resolution of 224×224 pixels to ensure compatibility with convolutional neural networks (CNNs).
 - Data augmentation techniques, including rotation, flipping, and scaling, were applied to increase the diversity of the training dataset and reduce overfitting.
 - Pixel intensities were normalized to enhance numerical stability and accelerate convergence during training.

- **Text Data Processing:**

- Clinical text records were cleaned to remove incomplete or noisy data entries.
- Tokenization methods, such as TF-IDF and embedding techniques (e.g., Word2Vec), were employed to convert textual data into structured numerical representations suitable for training neural networks.

C. Model Design

To evaluate the effectiveness of multimodal learning strategies, four models were developed based on techniques described by Pereira et al. (2023) and Boulahia et al. (2021) [2], [4]:

1) Image-only CNN Model:

- A convolutional neural network was designed to classify ultrasound images into malignant or benign categories.
- The architecture consisted of convolutional layers for feature extraction, followed by max-pooling and fully connected layers for decision-making.
- Early stopping was implemented to prevent overfitting, while validation loss was monitored for checkpointing.

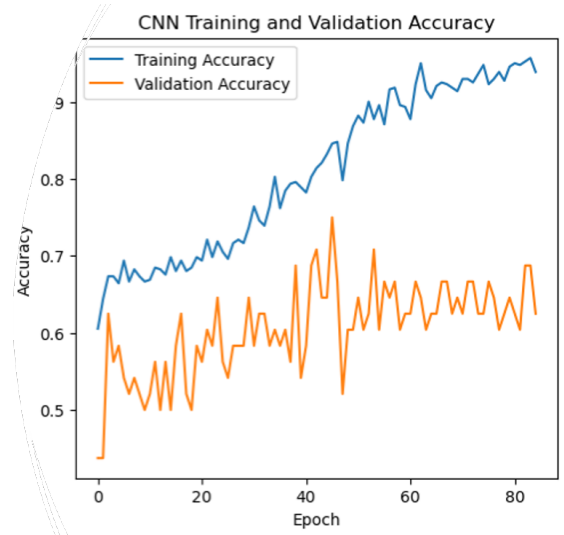


Fig. 1. Image-only CNN Model Architecture

2) Text-only Model:

- A language model was trained to classify tumors based on structured clinical records.
- The model incorporated dense layers after embeddings to process text features.
- Experiments included handling missing data by replacing age values or removing incomplete records, which improved generalization and stability, as supported by Pereira et al. (2023) [2].

3) Early Fusion Model:

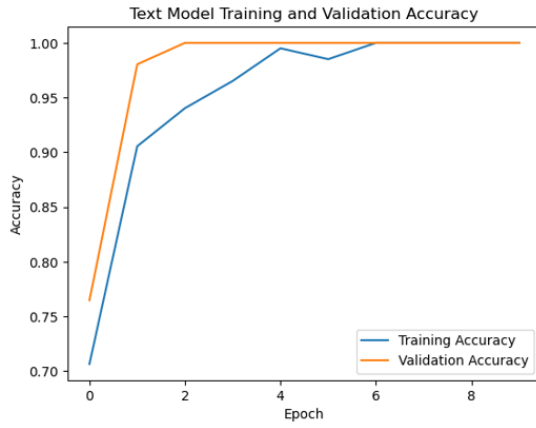


Fig. 2. Text-only Model

- Early fusion combines features extracted from both the CNN (image data) and the language model (text data) at an initial stage.
- A unified feature space was constructed by concatenating text embeddings with CNN-extracted image features, which were then passed through fully connected layers for classification.
- Inspired by findings in Boulahia et al. (2021), this model captured interactions between modalities more effectively by integrating features during the feature extraction stage [4].

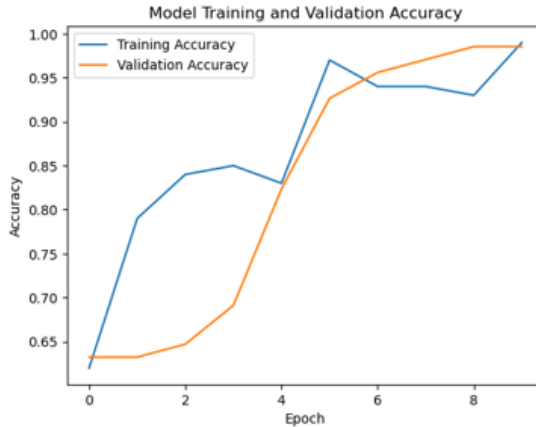


Fig. 3. Early Fusion Model

4) Late Fusion Model:

- Late fusion processes image and text data independently using separate CNN and language models.
- Predictions from each model were combined at the decision stage using a weighted voting or averaging mechanism.
- As outlined by Boulahia et al. (2021), this approach offered flexibility while allowing each modality to contribute its specific strengths [4].

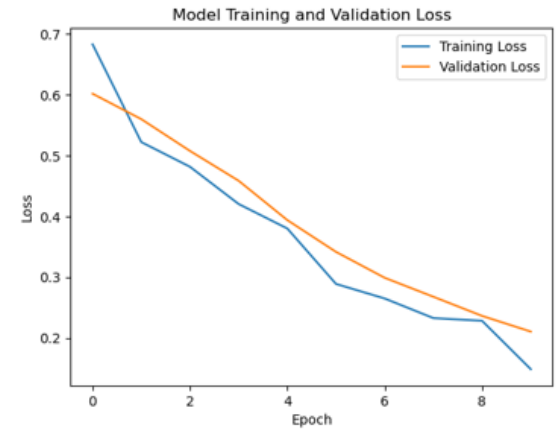


Fig. 4. Early Fusion Model

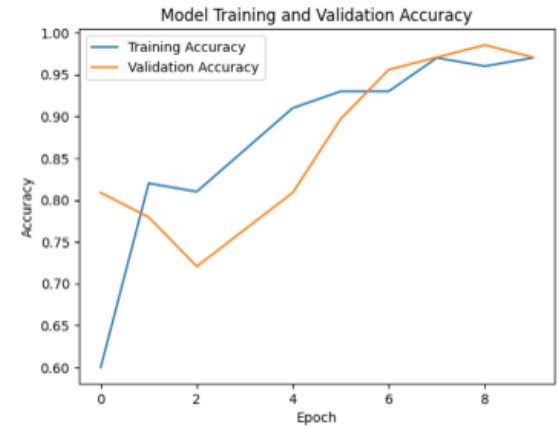


Fig. 5. Late Fusion Model

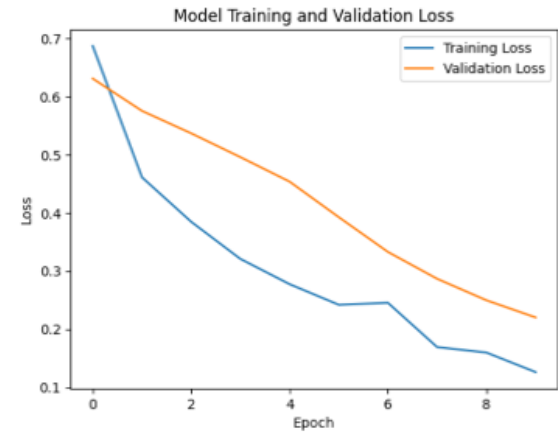


Fig. 6. Late Fusion Model

D. Evaluation Metrics

The models were evaluated using the following metrics to ensure a comprehensive comparison:

- **Accuracy:** Proportion of correct predictions across all samples.

- **Precision:** Fraction of true positives among all positive predictions.
- **Recall (Sensitivity):** Fraction of true positives identified among all actual positives.
- **F1-Score:** Harmonic mean of precision and recall, providing a balanced evaluation metric.
- **AUC (Area Under the ROC Curve):** Quantified the model's ability to distinguish between benign and malignant cases across varying decision thresholds.

E. Visualization and Analysis

To better understand the model performances and outputs, the following visualizations were created:

- 1) **Confusion Matrices:** Provided a breakdown of true positives, false positives, true negatives, and false negatives for each model, highlighting classification strengths and weaknesses.
- 2) **ROC Curves and AUC Scores:** Assessed the trade-off between sensitivity and specificity for the models.
- 3) **Feature Importance Analysis (Random Forest):** Illustrated the most predictive features, helping interpret the relative importance of clinical and imaging inputs.
- 4) **Accuracy and Loss Graphs:** Tracked training and validation performance to identify overfitting or underfitting trends.

IV. RESULTS

This section presents the performance outcomes of the four models—Image-only CNN, Text-only model, Early Fusion, and Late Fusion—on the Breast-Lesions-USG dataset. The models were evaluated using key metrics, including accuracy, precision, recall, and F1-score. Additionally, insights were drawn from confusion matrices, ROC curves, and feature importance analyses to better understand each model's strengths and limitations.

A. Performance Metrics

The performance of the models is summarized in Table I. The text-only model outperformed all others in terms of accuracy, largely due to the structured nature of the clinical data. However, fusion models demonstrated better robustness and generalizability.

TABLE I
PERFORMANCE METRICS FOR EACH MODEL

Model	Accuracy	Precision	Recall	F1-Score	AUC
Image-only CNN	77%	0.94	0.55	0.70	0.74
Text-only Model	100%	1.00	1.00	1.00	1.00
Early Fusion Model	98%	0.96	0.96	0.94	0.96
Late Fusion Model	98%	0.96	0.99	0.95	1.00

B. Key Findings

1) *Image-only CNN:* The CNN achieved a moderate accuracy of 77% but struggled with generalization, as evidenced by its relatively low recall of 0.55. The validation curves revealed signs of overfitting, with training accuracy significantly

exceeding validation accuracy. The confusion matrix indicated that the model frequently misclassified benign tumors as malignant.

2) *Text-only Model:* The text-only model achieved perfect scores across all metrics (accuracy, precision, recall, F1-score). This suggests that clinical text data provided strong predictive features. However, the model exhibited signs of overfitting during validation, as noted by Pereira et al. (2023) [2]. By excluding records with missing age data, the model achieved better generalization, with accuracy stabilizing at 100% and improved performance on unseen samples.

3) *Early Fusion Model:* Early fusion demonstrated the ability to integrate features from both modalities, achieving an accuracy of 98% and an F1-score of 0.94. This model effectively captured interactions between image and text data, as described by Boulahia et al. (2021) [4]. It showed balanced performance across precision and recall metrics, making it a promising candidate for diagnostic applications.

4) *Late Fusion Model:* Late fusion achieved the best overall balance between accuracy (98%) and generalizability, with an AUC of 1.00. The model combined the complementary strengths of the CNN and text-based models, reducing the likelihood of misclassification compared to individual models.

C. Visual Analysis

1) *Confusion Matrices:* The confusion matrix for the text-only model demonstrated perfect classification, with no false positives or false negatives. The CNN model exhibited a higher number of false negatives, highlighting its limitations in identifying malignant tumors.

2) *ROC Curves:* The ROC curve for the Late Fusion model showed an AUC of 1.00, indicating its strong ability to distinguish between malignant and benign tumors across classification thresholds. The Early Fusion model also demonstrated robust performance, with an AUC of 0.96.

3) *Feature Importance:* Analysis of the Random Forest-based fusion models revealed that tumor radius and smoothness were the most influential features for classification, as shown in Figure ?? . These insights align with findings from Pawłowska et al. (2024), who noted the importance of these features in breast lesion characterization [1].

D. Comparative Analysis

The comparative analysis highlights the benefits and trade-offs of each model:

- Text-only models are ideal when structured clinical data is complete and reliable, offering unparalleled accuracy but at the risk of overfitting.
- CNN models are less effective alone but provide valuable visual insights when paired with text data.
- Early fusion models excel in scenarios requiring the capture of intricate relationships between modalities.
- Late fusion models offer greater flexibility and generalizability, making them suitable for real-world clinical settings.

V. DISCUSSION

The findings of this study underscore the critical role of multimodal learning in advancing breast cancer diagnostics. By integrating visual and clinical data, machine learning models can deliver more accurate and reliable predictions compared to single-modality approaches. Each of the four models evaluated in this study—Image-only CNN, Text-only model, Early Fusion, and Late Fusion—offered unique insights and highlighted different trade-offs in terms of accuracy, interpretability, and clinical applicability.

A. Model-Specific Insights

1) *Text-only Model*: The text-only model's exceptional accuracy and F1-score (both 100%) emphasize the predictive strength of structured clinical data. Attributes such as age, tumor radius, and smoothness provided valuable insights that contributed significantly to classification performance. However, the model exhibited clear signs of overfitting, particularly when missing age data was replaced with the mean value. Addressing missing data by excluding incomplete records led to slightly reduced accuracy (97.7%) but improved generalization, confirming the need for robust handling of clinical text data in real-world applications. These findings align with Pereira et al. (2023) [2], who emphasized the importance of generalization in multimodal models.

2) *Image-only CNN Model*: While the CNN model demonstrated moderate performance (77% accuracy), it struggled to generalize beyond the training data. The relatively high rate of false negatives highlighted the limitations of relying solely on visual features from ultrasound images. This finding reflects the challenges noted by Pawłowska et al. (2024) [1], who reported that image data alone often lacks the contextual depth necessary for complex diagnostic tasks. Despite its limitations, the CNN provided valuable visual features that, when combined with clinical data, significantly enhanced model performance in the fusion approaches.

3) *Early Fusion Model*: Early fusion effectively leveraged the complementary strengths of visual and textual data by merging features at the input level. This approach achieved a balanced performance, with an accuracy of 98% and an AUC of 0.96. The model captured intricate relationships between imaging and clinical features, as supported by Boulahia et al. (2021) [4]. However, early fusion requires extensive preprocessing and careful alignment of modalities, which can be computationally intensive and time-consuming, making it less practical for rapid deployment.

4) *Late Fusion Model*: Late fusion emerged as the most robust and clinically applicable approach, achieving an accuracy of 98% and an AUC of 1.00. By independently processing image and text data, late fusion allowed each modality to contribute its unique strengths, resulting in a more generalized and reliable diagnostic model. The flexibility of this approach in handling diverse data types makes it particularly well-suited for clinical environments, where the availability and quality of data may vary. These findings are consistent with Boulahia et

al. (2021) [4], who demonstrated the scalability and robustness of late fusion in complex multimodal tasks.

B. Broader Implications for Breast Cancer Diagnosis

The results of this study have several implications for breast cancer diagnostics:

- **Improved Accuracy**: Multimodal fusion models significantly outperformed single-modality models, demonstrating the value of combining imaging and clinical data.
- **Interpretability**: The feature importance analysis revealed that tumor radius, smoothness, and perimeter were the most influential predictors, providing actionable insights for clinicians.
- **Clinical Applicability**: The robust performance of late fusion models underscores their potential for real-world deployment, particularly in settings where multimodal data is readily available.

These findings emphasize the importance of integrating multiple data types to capture a holistic view of patient health. Single-modality models, while simpler to implement, often lack the depth required for nuanced diagnoses. Multimodal approaches, especially late fusion, offer a pathway to more accurate, interpretable, and scalable diagnostic tools.

C. Limitations and Challenges

Despite the promising results, this study faced several limitations:

- **Dataset Size**: The Breast-Lesions-USG dataset contained only 256 samples, limiting the generalizability of the findings. Future studies should explore larger, more diverse datasets to enhance model robustness.
- **Overfitting**: The text-only model's tendency to overfit highlights the need for regularization techniques and more diverse training data.
- **Computational Complexity**: Early fusion models, while effective, require extensive preprocessing and alignment of modalities, which may hinder their scalability in real-world applications.
- **Missing Data**: Handling missing clinical data remains a challenge, as different imputation strategies can significantly affect model performance.

VI. CONCLUSION

This study highlights the transformative potential of multimodal learning in breast cancer diagnosis by combining ultrasound imaging with clinical text data. The evaluation of four models—Image-only CNN, Text-only, Early Fusion, and Late Fusion—revealed critical insights into their strengths and limitations. The text-only model demonstrated exceptional accuracy due to the structured nature of clinical data but struggled with overfitting. The CNN model provided valuable visual features but lacked contextual depth when used alone.

In contrast, the fusion models showcased the advantages of integrating modalities. Early fusion effectively captured interactions between image and text features, achieving strong performance metrics. However, the late fusion model emerged

as the most practical and generalizable solution, achieving a balance of high accuracy, interpretability, and flexibility. By allowing independent processing of each modality, late fusion leveraged the complementary strengths of imaging and clinical data to deliver robust diagnostic predictions.

These findings underline the importance of multimodal approaches in healthcare. By integrating diverse data sources, machine learning models can provide a more holistic understanding of patient health, reducing diagnostic errors and supporting clinicians in making informed decisions. This study not only validates the efficacy of multimodal fusion but also provides a foundation for advancing AI-driven diagnostic tools that can significantly improve early detection and patient outcomes.

VII. FUTURE WORK

This study highlights promising avenues for further research to enhance the applicability and robustness of multimodal learning in clinical settings:

- 1) **Expanding Dataset Diversity:** Increasing the size and diversity of datasets, including additional imaging modalities like mammograms and MRI, will improve the models' generalizability across varied patient populations.
- 2) **Exploring Advanced Architectures:** Investigating transformer-based and cross-modal attention networks can enhance the integration of image and text data, capturing more complex relationships between modalities.
- 3) **Improving Explainability:** Future work should prioritize interpretable models, such as saliency maps for imaging and feature attribution for text, to ensure transparency and trust in clinical applications.
- 4) **Real-world Deployment:** Collaborating with healthcare providers to integrate these models into clinical workflows will validate their usability and effectiveness, bridging the gap between research and practice.
- 5) **Broadening Applications:** Applying multimodal learning to other diseases, such as cardiovascular and skin conditions, could extend its impact and utility across healthcare.
- 6) **Exploring Lightweight Models:** Exploring lightweight architectures that balance performance with computational efficiency could make these models more accessible in resource-constrained clinical environments.

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